

Control work 2, and the quarter review

Parallelism in space — lines, planes, projection and area — in one paper, then the map.

LESSON 31–32

2 × 40 MIN

GEOMETRY 10, NAZORAT ISHI 2

IGCSE E4 REVIEW

LESSON CLOCK

3

40'

12'

22'

10'

3

Instructions	3 min
The paper	40 min
Answers	12 min
Rewrite	22 min
Concept map	10 min
Targets	3 min

LEARNING OBJECTIVES

- Apply the whole quarter in one assessment.
- Quote a test or a theorem by name in every proof.
- Classify each lost mark and rewrite the solution.
- Build a concept map of parallelism in space.

TEXTBOOK

National	pp. 143–146
Cambridge	Core & Extended, pp. 313–316
Workbook	Control paper G2

01

Explanation

The paper — 25 marks, 40 minutes

Q	TASK	MARKS	FROM
1	List every edge of a cube skew to CD , and name one common perpendicular	4	L21–22
2	Prove $B_1C_1 \parallel ABCD$, quoting the test	4	L23–24
3	Prove $ABB_1A_1 \parallel DCC_1D_1$, quoting the test	4	L25–26
4	Give a counterexample showing perpendicularity in space is not transitive	3	L19–20
5	List four properties preserved by parallel projection and two that are not	4	L27–28
6	Cube of side 6: find the area of the section through A, C, B_1	6	L29–30

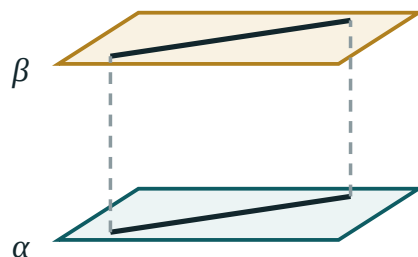
PROOFS ARE MARKED ON THE REASON

Q2 and Q3 each give 3 of their 4 marks for naming the test and verifying its conditions. A correct conclusion with no reason scores one.

The concept map

Six boxes, links as sentences:

- **two lines** → **three cases** — “parallel, meeting, or skew”
- **skew** → **common perpendicular** — “exactly one, and it is the distance”
- **line and plane** → **the test** — “parallel to *some* line of the plane”
- **two planes** → **the test** — “two *intersecting* lines, respectively parallel”
- **parallel planes** → **equal segments** — “why a prism’s lateral edges are equal”
- **projection** → **what a drawing may be trusted for** — “structure, never size”



$\alpha \parallel \beta$ – a third plane cuts them in parallel lines

The picture behind two of the six boxes.

Looking forward

Quarter III is perpendicularity — the theorem of the three perpendiculars, the distance from a point to a plane, and the dihedral angle. Every proof will use a parallelism result from this quarter, and every construction the projection rules of Lesson 27–28.

ONE HABIT TO CARRY FORWARD

Name the test, then verify each of its conditions in turn. Two of the four marks in every proof question this quarter were for exactly that.

02 Worked examples

EXAMPLE 1 Model answer, Q2: prove $B_1C_1 \parallel ABCD$.

- 1 $B_1C_1 \not\subset ABCD$
Condition 1 of the test.
- 2 $B_1C_1 \parallel BC$
Opposite sides of square BCC_1B_1 .
- 3 $BC \subset ABCD$
Condition 2.
- 4 By the test for a line parallel to a plane.

ANSWER

$$B_1C_1 \parallel ABCD$$

EXAMPLE 2 Model answer, Q3: prove $ABB_1A_1 \parallel DCC_1D_1$.

- 1 $AB \parallel DC$ and $AA_1 \parallel DD_1$.
- 2 AB and AA_1 meet at A .
Two intersecting lines.
- 3 DC and DD_1 lie in DCC_1D_1 .
- 4 By the test for parallel planes.

ANSWER

$$ABB_1A_1 \parallel DCC_1D_1$$

EXAMPLE 3 Model answer, Q6: cube of side 6, section through A, C, B₁.

- ① Each side is a face diagonal, $6\sqrt{2}$.

By rule R1 three times.

- ② Equilateral, so every angle is 60° .

③ $S = \frac{1}{2} \cdot 72 \cdot \sin 60^\circ$

④ $= 18\sqrt{3} \approx 31.2$

ANSWER

$$18\sqrt{3} \approx 31.2$$

03 Interactive model

Work Q2 and Q3 on the board, writing the test out in full before applying it.

The quarter in ten questions

Parallel lines in space must be:

non-meeting

coplanar and non-meeting

perpendicular

equal

Perpendicularity in space is:

transitive

not transitive

symmetric only

undefined

How many edges of a cube are skew to a given edge?

2

3

4

6

Two skew lines have how many common perpendiculars?

0

1

2

∞

The test for $\ell \parallel \alpha$ needs:

every line of α

one line of α

a perpendicular

two planes

The test for parallel planes needs:

one parallel line

two intersecting parallel lines

three points

a common point

A third plane cuts two parallel planes in:

parallel lines

perpendicular lines

a point

nothing

Parallel projection preserves:

lengths

angles

midpoints

areas

The image of a circle is:

a circle

an ellipse

a line

a parabola

Cube side 6: area of the section ACB_1 :

18

$18\sqrt{3}$

36

$36\sqrt{2}$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Test (criterion)	Alomat	Признак
Justification	Asoslash	Обоснование
Concept map	Tushunchalar xaritasi	Карта понятий
Revision	Takrorlash	Повторение
Self-assessment	O'z-o'zini baholash	Самооценка
Target	Maqsad	Цель
Counterexample	Qarshi misol	Контрпример

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A proof scores most of its marks for:

the conclusion

naming and checking the test

neatness

speed

The test for parallel planes fails with:

two intersecting lines

one line only

three lines

a perpendicular

A drawing may be trusted for:

lengths

angles

structure

areas

Quarter III begins with:

- volume
- perpendicularity in space
- vectors
- probability

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Edges skew to CD in a cube

ANSWER $AA_1, BB_1, A_1D_1, B_1C_1$

2. Is $B_1C_1 \parallel ABCD$?

ANSWER yes

3. Is $ABB_1A_1 \parallel DCC_1D_1$?

ANSWER yes

4. Is perpendicularity transitive in space?

ANSWER no

5. Does projection preserve midpoints?

ANSWER yes

6. Does projection preserve angles?

ANSWER no

7. Area with $a = 6, b = 8, C = 30^\circ$

ANSWER 12

Medium · the standard question

7 PROBLEMS

1. Prove $B_1C_1 \parallel ABCD$

ANSWER Test: $B_1C_1 \parallel BC \subset ABCD$

2. Prove $ABB_1A_1 \parallel DCC_1D_1$

ANSWER Two intersecting parallel pairs

3. Common perpendicular of CD and AA_1

ANSWER AD

4. Cube side 6: area of section ACB_1

ANSWER $18\sqrt{3}$

5. Cube side 6: distance between AB and CC_1

ANSWER 6

6. Image of a square under projection

ANSWER a parallelogram

7. Angle between AB and DC_1 in a cube

ANSWER 45°

Hard · stretch and reason

7 PROBLEMS

1. Cube side a : distance between AB and B_1D_1

ANSWER $\frac{a}{\sqrt{2}}$

2. Prove segments of parallel lines between parallel planes are equal

ANSWER They bound a parallelogram

3. Cube side 6: area of the midpoint hexagonal section

ANSWER $27\sqrt{3} \approx 46.8$

4. Prove: if $\ell \parallel \alpha$ and $\ell \parallel \beta$ with $\alpha \cap \beta = m$ then $\ell \parallel m$

ANSWER Converse theorem in each plane

5. Tetrahedron: prove the midpoint plane is parallel to a face

ANSWER Two midlines, two directions

6. Give two planes parallel to the same line that intersect

ANSWER Two walls and a vertical line

7. Heron: sides 8, 11, 13

ANSWER ≈ 43.8

07 Homework

Homework — 4 tasks

Bring the concept map to the first lesson of Quarter III.

1. Rewrite in full every control-work question that lost a mark, naming the test used.
2. Finish the concept map with all six links written as sentences.
3. Cube of side 10: find the area of the section through A , C , B_1 and the distance between AB and CC_1 .

4. Write your target for Quarter III in one checkable sentence, and date it.